

Assignment 4

30 Marks

Deadline: Aug 31, 2026 11:59 PM (Monday)

Question 01

Suppose you have a single instruction;

Add X21, X22, X23

Register read = 20s

Instruction Fetch = 10s

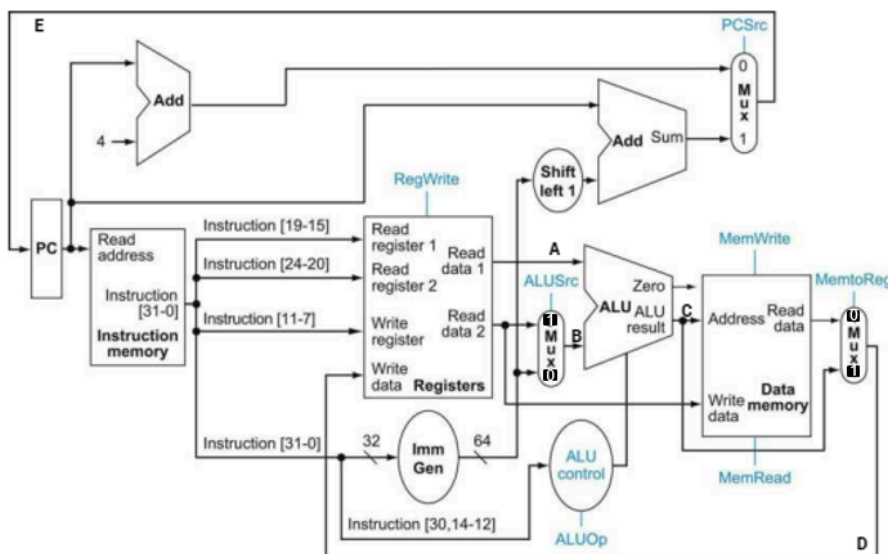
Memory Operation = 40s

Register write = 20s

ALU Operation = 40s

Suppose you have a pipelined setup and a single cycle setup,
Which one will run the instruction faster? [5]

Question 02



Write the control signals for the following instruction:
Ld X5, 10(X21);

- i. ALU Src;
- ii. RegWrite;
- iii. MemRead;
- iv. MemWrite;
- v. MemtoReg. [5]

Question 03

Draw a simplified datapath with control unit which can execute the following instruction,

- i) BLT X9, X14, Label2

Question 04

Why do we keep instruction and data memory separate? [5]

Question 05

Instruction
SD X22, 12(X21)
Add X21, X22, X23
LD X23, 12(X21)
Sub X11, X25, X16
Addi X21, X23, 7
BEQ X21, X22, L1
LD X22, 0(X21)

Will any data hazards occur during the execution of this instruction set?
If so, then mitigate the hazard using stall + forwarding + code
scheduling techniques where applicable and calculate average CPI. [10]